

Transet:	x_1	x_2	y
	1	1	2
	3	1	4

Test set: $x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$h_{\theta}(x) = \theta^T \phi(x) \quad \theta \in \mathbb{R}^p, \phi(x) \in \mathbb{R}^p$$

$$h_{\theta}(x) = \sum_{i=1}^n \beta_i k(x^{(i)}, x)$$

Let us assume:

$$\bullet \beta_1 = 1, \beta_2 = 1$$

$\bullet$ Kernel = Radial Basis Function (RBF)

$$k(x, z) = \exp(-\|x - z\|^2)$$

where,

$$\|x - z\|^2 = (x_1 - z_1)^2 + (x_2 - z_2)^2$$

$$\begin{aligned} h_{\theta}(x) &= \sum_{i=1}^n \beta_i k(x^{(i)}, x) \\ &= \beta_1 k(x^{(1)}, x) + \beta_2 k(x^{(2)}, x) \\ &= \beta_1 \exp(-\|x^{(1)} - x\|^2) + \beta_2 \exp(-\|x^{(2)} - x\|^2) \end{aligned}$$

$$(i) \|x^{(1)} - x\|^2$$

$$\begin{aligned} &= (x_1^{(1)} - x_1)^2 + (x_2^{(1)} - x_2)^2 \\ &= (1 - 2)^2 + (1 - 1)^2 \quad \xrightarrow{\text{from test data}} \\ &= (-1)^2 + (0)^2 = 1 \end{aligned}$$

$$\begin{aligned} (ii) \|x^{(2)} - x\|^2 &= (x_1^{(2)} - x_1)^2 + (x_2^{(2)} - x_2)^2 \\ &= (3 - 2)^2 + (1 - 1)^2 \\ &= (1)^2 + (0)^2 = 1 \end{aligned}$$

$$\begin{aligned} h_{\theta}(x) &= 1 \exp(-1) + 1 \exp(-1) \\ &= 0.367 + 0.367 \end{aligned}$$

$$h_{\theta}(x) = 0.735 //$$